

10.5

Surface Integral (integral over surfaces)

01

Representation of surface

Surface S in xyz -space are

$$\boxed{z = f(x, y)} \quad (1)$$

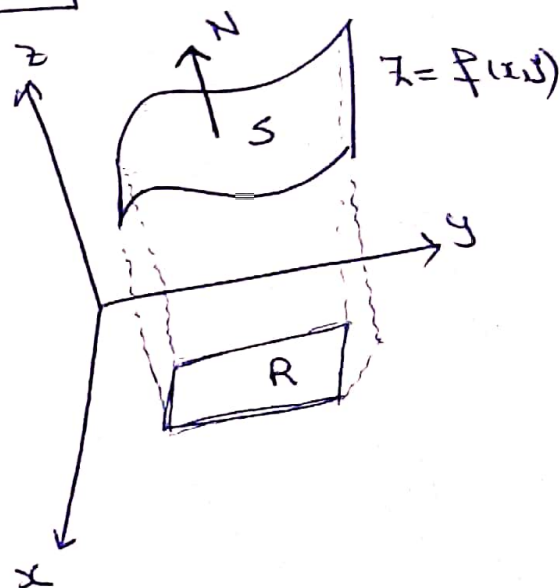
OR

$$z - f(x, y) = 0$$

$$\boxed{g(x, y, z) = 0} \quad (2)$$

Solution steps

To define a surface integral, we take a surface S , given by parametric equation,



$$\begin{aligned} \vec{r}(u, v) &= [x(u, v), y(u, v), z(u, v)] \\ &= x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k} \end{aligned} \quad (1)$$

where u, v define in $\mathbb{R}$

$$\vec{N} = \vec{r}_u \times \vec{r}_v \quad \left(\begin{array}{l} \text{unit normal vector to} \\ \text{surface} \end{array} \right) \quad (2)$$

$$\hat{n} = \frac{\vec{N}}{|\vec{N}|} \quad (\text{unit normal vector})$$

Q2

Q3 Now, surface integral over S is

$$\text{Flux} = \left| \int_S \vec{F} \cdot \hat{n} \, dS = \int_R \left[\vec{F}(x(u,v)) \cdot \vec{N}(u,v) \right] du \, dv \right|$$

↳ (3)

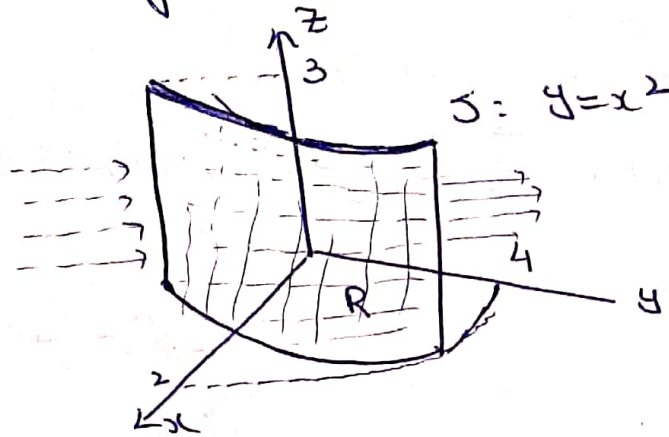
Ex Flux through a surface

L03

Compute the flux of vector through

parabolic cylinder $S: y = x^2, 0 \leq x \leq 2$
 $0 \leq z \leq 3$ (1)

if the velocity vector is $\vec{V} = \vec{F} = 3z^2 \hat{i} + 6\hat{j} + 6xz\hat{k}$
 (2)



Sol
01

Parametric equations

Let $x = u, y = v$
 (3) (4)

$$\begin{cases} 0 \leq u \leq 2 \\ 0 \leq v \leq 3 \end{cases} (*)$$

$$(1) \Rightarrow y = x^2 = u^2 \quad (5)$$

$$\vec{r}(u, v) = [u, u^2, v] = u\hat{i} + u^2\hat{j} + v\hat{k} \quad (6)$$

02

$$\vec{r}_u = \hat{i} + 2u\hat{j} + 0\hat{k} \quad (7)$$

$$\vec{r}_v = 0\hat{i} + 0\hat{j} + \hat{k} \quad (8)$$

$$\vec{N} = \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2u & 0 \\ 0 & 0 & 1 \end{vmatrix} = 2u\hat{i} - \hat{j} + 0\hat{k} = [2u, -1, 0] \quad (9)$$

$$\text{Q3} \quad \vec{F} = [3z^2, 6, 6xz]$$

$$x \rightarrow x(u,v)$$

$$y \rightarrow y(u,v)$$

$$z \rightarrow z(u,v)$$

$$(-\text{from } 6)$$

$$\vec{F}(x(u,v)) = [3v^2, 6, 6uv]$$

(10)

$$\text{Q4} \quad \vec{F}(x(u,v)) \cdot \vec{N} = \left(3v^2 \hat{i} + 6 \hat{j} + 6uv \hat{k} \right) \cdot \left(2u \hat{i} - \hat{j} + v \hat{k} \right)$$

$$= 6uv^2 - 6 \quad (11)$$

$$\text{Q5} \quad \text{Flux} = \int_S \vec{F} \cdot \vec{n} \, dS = \int_{v=0}^{v=3} \int_{u=0}^{u=2} (6uv^2 - 6) \, du \, dv$$

$$= \int_0^3 \left(6 \frac{u^2}{2} v^2 - 6u \right) \Big|_{u=0}^{u=2} \, dv$$

$$= \int_0^3 (12v^2 - 12) \, dv = 72 \, \text{m}^3/\text{sec}$$

Exercise Determine Flux Integral

(05)

$$\iint_S \vec{F} \cdot \vec{n} \, dS$$

~~Q1~~

① $\vec{F} = [2x, 5y, 0]$

$$\vec{r}(u, v) = [u, v, 4u + 3v]$$

$$0 \leq u \leq 1, \quad -2 \leq v \leq 2$$

02

$$\vec{F} = [x - z, y - x, z - y]$$

$$\vec{r}(u, v) = [u \cos v, u \sin v, u]$$

$$0 \leq u \leq 3, \quad 0 \leq v \leq \pi$$

03

$$\vec{F} = [x, y, z]$$

$$\vec{r}(u, v) = [u \cos v, u \sin v, u^2]$$

$$0 \leq u \leq 4, \quad -\pi \leq v \leq \pi$$